

## Element B

### *Documentation and Analysis of Prior Design Attempts*

Scientists have previously made numerous attempts to create various hydroponic machines. These experiments have revealed that the performance of these machines vary depending on the type of plant being grown. In order to improve efficiency, scientists have developed different versions of these machines, altering their position, size, and other aspects. Some designs have even been made more compact to optimize space utilization. Our team has conducted extensive experiments with different design sketches, aiming to maximize space usage and create visually appealing hydroponic machines for classroom settings. After careful consideration, the team has narrowed down our options to sharp-edged designs and have also contemplated incorporating a trellis to support viny plants. Our designs have been inspired by successful models that have effectively grown various types of plants, both vertically and horizontally. While our focus is on classroom hydroponic systems, it is worth nothing that many hydroponic machines in laboratories are used for testing purposes or even for food production. Lettuce, in particular, has proven to thrive exceptionally well in hydroponic systems. Our team has taken into account various factors that have caused previous systems to fail, ensuring that our design addresses these



Small Hydroponic Unit found in Classroom used by Homegrown Heroes Club to Grow Herbs to Dry



Large Hydroponic Unit found in Classroom used by Homegrown Heroes Club to Grow Lettuce

challenges effectively. A source labeled as a “beginners guide” gave the following list of the most important components to focus on. Water is crucial for hydroponic systems as it hydrates plants and provides a home where they can thrive. Without oxygen, plants will drown in the water and need additional O<sub>2</sub> from an air pump to regulate the oxygen level. Root bases like peat moss, rock wool, clay pellets, and rice hulls provide support and protection to the roots. Nutrient solutions must be added to the water for plants to absorb essential plant nutrients like nitrogen, phosphorus, and potassium. Full spectrum LED lights are necessary for photosynthesis and plant growth ([SOURCE](#)). Another source which helped us narrow down our design choices by showing us how different designs had different helpful aspects, hydroponic design involves using a growing container, a sump tank, and a pump. Different types of hydroponic systems include deep water culture, nutrient film technique, vertical tower systems, and aeroponic systems. When choosing a technique, growers should consider factors such as location, crop temperature, and facility. The Kratky method is a simple method that combines the sump with the growing container. When planning a commercial farm, it is important to consider labor costs and the size of the farm ([SOURCE](#)). Through the two sources the team has provided in this document it is

seen that there are many solutions to the problem, and many places where problems can arise from our team focusing on only one aspect.